NITHIN KIDANGAZHIATH MANA

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SUMMARY

VLSI Design and Verification-focused ECE graduate from Johns Hopkins University with experience in FPGA, and ASIC design and functional verification. Skilled in Verilog and SystemVerilog languages with demonstrated work in hardware acceleration of State Space Models and in-memory computing. Seeking to contribute to the design and development of scalable, high-performance digital systems and next-generation computing solutions.

EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, MD
MSE in Electrical and Computer Engineering | Graduated - 05/2026 | 3.6/4.0

VELLORE INSTITUTE OF TECHNOLOGY, VIT | Tamil Nadu, India
B.Tech in Electrical and Electronics Engineering |Graduated - 07/2024 | 3.7/4.0

WORK EXPERIENCE

Teaching Assistant: First-Year ECE Design | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026

- Mentored 30+ students over weekly lab sections and office hours, teaching circuit fundamentals and electrical design concepts.
- Graded homework, quizzes, lab reports, and communicated technical feedback.

ML Hardware Architect - Research| Johns Hopkins University | Baltimore, MD | May 2025 - January 2026

- Led **Verilog RTL Design and Verification** of a **low-power** 8-bit fixed-point quantized Vision Mamba **FPGA accelerator** for MedMNIST classification at the edge using **AMD Vivado 2024.1** (2).
- Identified key areas of **latency and performance bottlenecks** and delivered **9 times reduction in latency** (compared to baseline) via channel-level architecture parallelization of the SSM branch.
- Achieved design implementation within hardware and resource utilization constraints for **Kintex-7 XC7K325T FPGA**.
- **Published in IEEE Transactions on Biomedical Circuits and Systems**

Technical R&D Intern | STMicroelectronics | Uttar Pradesh, India | January 2024 - June 2024

- Spearheaded **RTL Design and Verification** of a 6T SRAM-based Analog In-Memory Compute architecture using **Simulink-based Verilog auto-generation** and **Cadence Xcelium simulation tool**.
- Developed a noise-aware hardware validation framework based on LFSRs to model ±0.2% bit-cell noise and ±1 feature-level perturbations, ensuring robustness in mixed-signal operation.

VLSI Design Intern | Maven Silicon Pvt. Ltd | Karnataka, India | May 2023 - July 2023

- Conducted **Verilog RTL Design and Verification** for an **AMBA AHB-to-APB bridge** supporting INCR4 and WRAP4 burst transfers, address pipelining, and FSM-controlled APB transfers using **Intel Quartus Prime software**.

TECHNICAL PROJECTS

Investigating State-Space ML Models for Memory-Efficient 3D Brain Tumor Segmentation| Johns Hopkins University | Baltimore, MD | January 2026 – May 2026 |

- Replaced shifted-window Transformer encoder blocks in **Swin-UNETR** with **Vision Mamba (SSM) modules** to evaluate state-space models as scalable alternatives for **3D medical image segmentation using PyTorch/Python** (1).
- Achieved 20% reduction in **peak GPU training memory,** 46.6% reduction in **peak inference memory, and comparable Dice and Hausdorff Distance** scores relative to Swin-UNETR baseline.

Reproducing and Extending Proximal Policy Optimization in Continuous-Control Tasks | Johns Hopkins University | Baltimore, MD | August 2025 – December 2026

- Pioneered an **extension to PPO** that incorporated **state-aware reset sampling**, systematically investigating its impact on exploration dynamics, policy stability, and reward convergence across continuous-control environments.
- Validated and stress-tested PPO performance on **MuJoCo benchmarks** through large-scale experimental campaigns, reproducing key findings from the seminal PPO study.

Design of 8-Bit & Binary Neural Network Accelerators| Johns Hopkins University | Baltimore, MD | August - December 2024

- Performed **Verilog RTL Design and Verification using Cadence Xcelium** of 8-bit quantized and binarized neural network ASIC accelerators for image classification**, logic synthesis using Cadence Genus**, demonstrating an ~8% reduction in combinational logic in the BNN design with equal area.
- Undertook **place-and-route using Cadence Innovus**, and power distribution analysis across combinational, sequential, memory, and IO components using post-layout PPA reports.

CERTIFICATIONS

- Cadence SystemVerilog for Design and Verification v25.03 Certification (Certified Feb 2026)
  Integrated and verified a **VeriRISC CPU in SystemVerilog**, validating instruction execution, ALU functionality, memory operations, and program flow using directed diagnostic tests and a Fibonacci benchmark program.

- **Cadence SystemVerilog Assertions v5.2 Certification (Certified March 2026)**
  Architected **assertion-based verification for an SPI interface**, capturing protocol sequences, transaction headers, suspend handling, illegal transaction detection, and coverage-driven validation using SVA.

- **Cadence Essential SystemVerilog for UVM v1.2.5rev3 Certification (Certified June 2026)**
  Developed **Verification Components** to test a **4-port switch design**, covering **constrained-random** verification, virtual interfaces, sequencer, driver, and monitor subclasses through **Object-Oriented Programming** principles.

- **Cadence SystemVerilog Accelerated Verification with UVM v1.2.5rev2 (Anticipated Certification June 2026)**
  Developing a **UVM verification environment for a YAPP packet-router DUT**, incorporating constrained-random stimulus generation, reusable UVCs, virtual interfaces, TLM scoreboarding, and functional coverage.

- The Ultimate C++ Series Course **(Online Certification)**

RELEVANT COURSEWORK

- CAD Design of Digital VLSI Systems (Graduate Coursework)
- FPGA Synthesis Lab (Graduate Coursework)
- Foundations of Reinforcement Learning (Graduate Coursework)
- VLSI Design (Undergraduate Coursework)
- Computer Architecture and Organization (Undergraduate Coursework)
- Object-Oriented Programming with C++ (Undergraduate Coursework)